EXPERIMENT - 5

Aim:-

Using the Employee schema, demonstrate:

- SQL View - Department Salary Summary
- SQL View - Employee Hierarchy
- Updatability of Views (simple view, CHECK OPTION view, aggregate view)
- Recursive CTE to display Reporting Chains

Source Code:-

ADD MANAGER COLUMN (SETUP FOR HIERARCHY)

```
ALTER TABLE Employee
ADD COLUMN manager_id INT NULL;

UPDATE Employee
SET manager_id = NULL
WHERE emp_id = 1;

UPDATE Employee
SET manager_id = FLOOR((emp_id - 2) / 5) + 1
WHERE emp_id > 1;

ALTER TABLE Employee
ADD CONSTRAINT fk_emp_manager
FOREIGN KEY (manager_id) REFERENCES Employee(emp_id);

SELECT emp_id, emp_name, manager_id
FROM Employee;
```

VIEW 1 - DEPARTMENT SALARY SUMMARY

```
CREATE OR REPLACE VIEW dept_salary_summary AS
SELECT
   d.dept_id,
   d.dept_name,
   COUNT(e.emp_id) AS total_employees,
   SUM(e.salary) AS total_salary,
   ROUND(AVG(e.salary), 2) AS avg_salary,
   MIN(e.salary) AS min_salary,
   MAX(e.salary) AS max_salary
FROM Department d
LEFT JOIN Employee e
ON d.dept_id = e.dept_id
GROUP BY d.dept_id, d.dept_name;

SELECT * FROM dept_salary_summary;
```

VIEW 2 - EMPLOYEE HIERARCHY

```
CREATE OR REPLACE VIEW emp_hierarchy AS
SELECT
   e.emp_id,
   e.emp_name,
   d.dept_name,
   e.manager_id,
   m.emp_name AS manager_name
FROM Employee e
LEFT JOIN Employee m
ON e.manager_id = m.emp_id
LEFT JOIN Department d
ON e.dept_id = d.dept_id;

SELECT * FROM emp_hierarchy
ORDER BY manager_id, emp_id;
```

UPDATABILITY TEST 1 - SIMPLE VIEW (UPDATABLE)

```
CREATE OR REPLACE VIEW emp_basic_view AS
SELECT emp_id, emp_name, dept_id, salary
FROM Employee;

UPDATE emp_basic_view
SET salary = salary + 1000
WHERE emp_id = 1;

SELECT emp_id, emp_name, salary
FROM Employee
WHERE emp_id = 1;

UPDATE emp_basic_view
SET salary = salary - 1000
WHERE emp_id = 1;
```

UPDATABILITY TEST 2 - VIEW WITH CHECK OPTION

```
CREATE OR REPLACE VIEW sciences_emp_view AS
SELECT emp_id, emp_name, dept_id, salary
FROM Employee
WHERE dept_id = 2
WITH CHECK OPTION;

-- Allowed: row still satisfies dept_id = 2
UPDATE sciences_emp_view
SET salary = salary + 500
WHERE emp_id = 2;

-- Rejected: new row would violate dept_id = 2
UPDATE sciences_emp_view
SET dept_id = 1
WHERE emp_id = 2;
```

```
-- Revert the valid change
UPDATE sciences_emp_view
SET salary = salary - 500
WHERE emp_id = 2;
```

UPDATABILITY TEST 3 - AGGREGATE VIEW (NOT UPDATABLE)

```
UPDATE dept_salary_summary
SET avg_salary = 100000
WHERE dept_id = 1;
```

CHECK UPDATABLE STATUS OF ALL VIEWS

```
SELECT
   table_name AS view_name,
   is_updatable
FROM information_schema.views
WHERE table_schema = DATABASE();
```

RECURSIVE CTE - REPORTING CHAIN (TOP DOWN)

```
WITH RECURSIVE reporting_chain AS
(
   SELECT
      emp_id,
      emp_name,
      manager_id,
      1 AS level,
      CAST(emp_name AS CHAR(500)) AS chain
   FROM Employee
   WHERE manager_id IS NULL

   UNION ALL

   SELECT
      e.emp_id,
      e.emp_name,
      e.manager_id,
      rc.level + 1,
      CONCAT(rc.chain, ' -> ', e.emp_name)
   FROM Employee e
   JOIN reporting_chain rc
   ON e.manager_id = rc.emp_id
)
SELECT emp_id, emp_name, level, chain
FROM reporting_chain
ORDER BY level, emp_id;
```

RECURSIVE CTE - REPORTING CHAIN OF ONE EMPLOYEE (BOTTOM UP)

```
WITH RECURSIVE upward_chain AS
(
   SELECT
```

```
        emp_id,
        emp_name,
        manager_id,
        0 AS steps_above
    FROM Employee
    WHERE emp_id = 27

    UNION ALL

    SELECT
        e.emp_id,
        e.emp_name,
        e.manager_id,
        uc.steps_above + 1
    FROM Employee e
    JOIN upward_chain uc
    ON e.emp_id = uc.manager_id
)
SELECT emp_id, emp_name, steps_above
FROM upward_chain
ORDER BY steps_above;
```

Output:-

ADD MANAGER COLUMN (SETUP FOR HIERARCHY)

UPDATE OK, 0 row(s) affected

UPDATE Employee SET manager_id = NULL WHERE emp_id = 1

UPDATE OK, 29 row(s) affected

UPDATE Employee SET manager_id = FLOOR((emp_id - 2) / 5) + 1 WHERE emp_id > 1

emp_id	emp_name	manager_id
1	Aarav Sharma	NULL
2	Ananya Singh	1
3	Priya Mehta	1
4	Karan Malhotra	1
5	Sneha Gupta	1
6	Rohan Verma	1
7	Rahul Joshi	2
8	Neha Kapoor	2
9	Varun Sethi	2
10	Aditi Chawla	2
11	Vikram Rao	2
12	Isha Patel	3
13	Kavya Shah	3
14	Mohit Bansal	3
15	Meera Nair	3

... showing first 15 of 30 rows

VIEW 1 - DEPARTMENT SALARY SUMMARY

dept_id	dept_name	total_employees	total_salary	avg_salary	min_salary	max_salary
1	Computer Science Engineering	7	439000.00	62714.29	58000.00	66000.00
2	Sciences	6	503000.00	83833.33	72000.00	95000.00
3	Law	6	441000.00	73500.00	66000.00	81000.00
4	Management	6	460000.00	76666.67	70000.00	85000.00
5	Commerce	5	339000.00	67800.00	65000.00	72000.00

VIEW 2 - EMPLOYEE HIERARCHY

emp_id	emp_name	dept_name	manager_id	manager_name
1	Aarav Sharma	Computer Science Engineering	NULL	NULL
2	Ananya Singh	Sciences	1	Aarav Sharma
3	Priya Mehta	Law	1	Aarav Sharma
4	Karan Malhotra	Management	1	Aarav Sharma
5	Sneha Gupta	Commerce	1	Aarav Sharma
6	Rohan Verma	Computer Science Engineering	1	Aarav Sharma
7	Rahul Joshi	Sciences	2	Ananya Singh
8	Neha Kapoor	Law	2	Ananya Singh
9	Varun Sethi	Management	2	Ananya Singh
10	Aditi Chawla	Commerce	2	Ananya Singh
11	Vikram Rao	Sciences	2	Ananya Singh
12	Isha Patel	Computer Science Engineering	3	Priya Mehta
13	Kavya Shah	Law	3	Priya Mehta
14	Mohit Bansal	Management	3	Priya Mehta
15	Meera Nair	Law	3	Priya Mehta

... showing first 15 of 30 rows

UPDATABILITY TEST 1 - SIMPLE VIEW (UPDATABLE)

UPDATE OK, 1 row(s) affected

UPDATE emp_basic_view SET salary = salary + 1000 WHERE emp_id = 1

emp_id	emp_name	salary
1	Aarav Sharma	66000.00

UPDATE OK, 1 row(s) affected

UPDATE emp_basic_view SET salary = salary - 1000 WHERE emp_id = 1

UPDATABILITY TEST 2 - VIEW WITH CHECK OPTION

UPDATE OK, 1 row(s) affected

UPDATE sciences_emp_view SET salary = salary + 500 WHERE emp_id = 2

ERROR 1369: CHECK OPTION failed `lab`.`sciences_emp_view`

UPDATE sciences_emp_view SET dept_id = 1 WHERE emp_id = 2

UPDATE OK, 1 row(s) affected

UPDATE sciences_emp_view SET salary = salary - 500 WHERE emp_id = 2

UPDATABILITY TEST 3 - AGGREGATE VIEW (NOT UPDATABLE)

ERROR 1288: The target table dept_salary_summary of the UPDATE is not updatable

UPDATE dept_salary_summary SET avg_salary = 100000 WHERE dept_id = 1

CHECK UPDATABLE STATUS OF ALL VIEWS

view_name	is_updatable
sciences_emp_view	YES
emp_hierarchy	YES
dept_salary_summary	NO
emp_basic_view	YES

RECURSIVE CTE - REPORTING CHAIN (TOP DOWN)

emp_id	emp_name	level	chain
1	Aarav Sharma	1	Aarav Sharma
2	Ananya Singh	2	Aarav Sharma -> Ananya Singh
3	Priya Mehta	2	Aarav Sharma -> Priya Mehta
4	Karan Malhotra	2	Aarav Sharma -> Karan Malhotra
5	Sneha Gupta	2	Aarav Sharma -> Sneha Gupta
6	Rohan Verma	2	Aarav Sharma -> Rohan Verma
7	Rahul Joshi	3	Aarav Sharma -> Ananya Singh -> Rahul Joshi
8	Neha Kapoor	3	Aarav Sharma -> Ananya Singh -> Neha Kapoor
9	Varun Sethi	3	Aarav Sharma -> Ananya Singh -> Varun Sethi
10	Aditi Chawla	3	Aarav Sharma -> Ananya Singh -> Aditi Chawla
11	Vikram Rao	3	Aarav Sharma -> Ananya Singh -> Vikram Rao
12	Isha Patel	3	Aarav Sharma -> Priya Mehta -> Isha Patel
13	Kavya Shah	3	Aarav Sharma -> Priya Mehta -> Kavya Shah
14	Mohit Bansal	3	Aarav Sharma -> Priya Mehta -> Mohit Bansal
15	Meera Nair	3	Aarav Sharma -> Priya Mehta -> Meera Nair

... showing first 15 of 30 rows

RECURSIVE CTE - REPORTING CHAIN OF ONE EMPLOYEE (BOTTOM UP)

emp_id	emp_name	steps_above
27	Tanya Roy	0
6	Rohan Verma	1
1	Aarav Sharma	2